INTERNATIONAL MONETARY FUND

TONGA

STAFF REPORT FOR THE 2023 ARTICLE IV CONSULTATION —DEBT SUSTAINABILITY ANALYSIS

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Approved By
Ranil Salgado, Bergljot Bjornson Barkbu (IMF), and Manuela Francisco and Lalita Moorty (IDA)

Prepared by the staffs of the International Monetary Fund and the International Development Association

Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	High
Overall risk of debt distress	High
Granularity in the risk rating	Sustainable
Application of judgment	Yes

The risk of debt distress rating for Tonga remains high.[1] Before the pandemic, Tonga's indebtedness had gradually declined from FY2015[2] to FY2019 owing to much-needed fiscal consolidation. Since 2020, however, the economy has been hit by multiple shocks, including the COVID-19 pandemic, tropical cyclone Harold in April 2020, and the Hunga Tonga–Hunga Ha'apai (HTHH) volcanic eruption and tsunami in January 2022. In response, the international community has stepped up financial support to Tonga through grants, concessional loans, and debt service suspensions. Against this backdrop, the public debt-to-GDP ratio in FY2023 is expected to rise slightly from pre-pandemic levels. The budget deficit is expected to rise sharply in the coming years due to a gradual decrease in grants under the current commitments and growing spending needs for sustainable development goals (SDG) and climate resilience. Debt service payments will spike in FY2024 due to scheduled repayment to the China Exim Bank. Under the baseline scenario, the present value (PV) of the external debt-to-GDP ratio is expected to breach the threshold starting in FY2033, while the PV of the public debt-to-GDP ratio is also expected to exceed the benchmark in the same year. The breach will continue after FY2033 as both external and public debt continue to rise until 2043. A tailored, large one-time natural disaster shock would result in a significant deterioration in debt sustainability. Long-term debt sustainability hinges on fiscal adjustments and continued donor grant inflows to finance the necessary spending for climate resilience and SDGs. To rebuild fiscal buffers and enhance resilience against shocks, measures for stronger revenue mobilization and expenditure rationalization will be essential, together with external donor support consistent with the historical trends.

[1] The Tonga Composite Indicator (CI) index, calculated based on the April 2023 World Economic Outlook (WEO) and the 2021 Country Policy and Institutional Assessment (CPIA), is at 3.06, indicating that Tonga's debt-carrying capacity remains strong.

[2] All the figures are computed using fiscal year beginning in July, e.g., FY2022 runs from July 1, 2021, to June 30, 2022.

PUBLIC DEBT COVERAGE

1. Tonga's public debt includes obligations of the central government and central bank. The central bank's debt is borrowed on behalf of the government. Local governments do not incur debt, nor do other entities in the general government. As of end-June 2023, other government-guaranteed debt was small, about 1.1 percent of GDP. Since the Debt Sustainability Analysis (DSA) coverage does not include debt of state-owned enterprises (guaranteed and non-guaranteed) due to data limitations,[3] an additional 2 percent of GDP is added to the contingent liability test. Contingent liabilities also include a standard 5 percent of GDP cost to the government of a financial crisis, which is above the existing stock of financial sector NPLs.[4] The DSA is conducted on a residency basis.

Tonga: Public Debt Coverage

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	X
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The general government, central bank		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2.0	
4	PPP	35 percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BACKGROUND ON DEBT

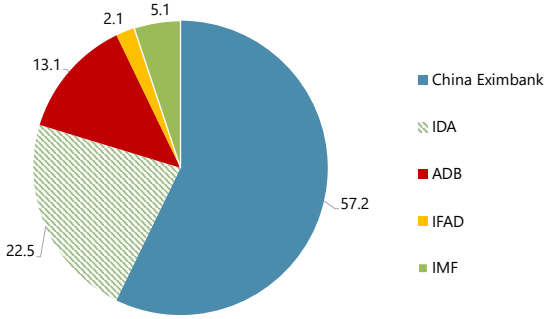
2. After years of decline, public debt level rose modestly since 2019. The public debt-to-GDP ratio declined from 51 percent at end-June 2015 to 41 percent at end-June 2019 (chart) due to fiscal consolidation since FY2016. The recent increase reflects new issuance of domestic debt and the two concessional emergency loans under the IMF Rapid Credit Facility (RCF). To contain unfavorable debt dynamics, the government's fiscal anchors (introduced in FY2018) limit total public external debt-to-GDP to a maximum of 50 percent. Tonga has not contracted any external non-concessional loans since FY2011, in line with their debt management strategy.

[3] The public debt data shared by the authorities do not include any guaranteed debt, although based on the FY2024 budget document, government guaranteed debt is projected to be about 1.1 percent of GDP by end-2023. The government is working to include guaranteed debt in their public debt data, including through improving quality and timely reporting of debt bulletin.

[4] Tonga does not have arrears to external creditors.

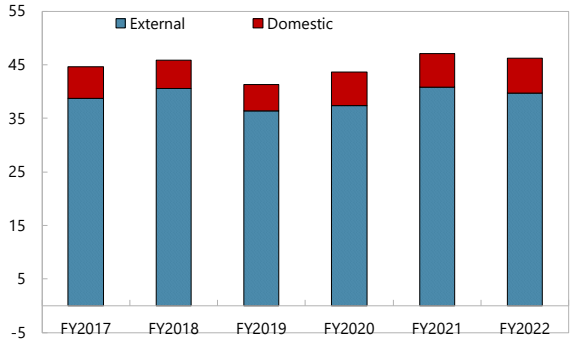
3. Tonga's debt obligations are largely external, and over half of its external public debt is to China, with a sharp spike in debt repayments due from FY2024 onwards. Total public and publicly guaranteed (PPG) external debt stood at USD185 million (about 38 percent of GDP) as of end-June 2022, accounting for 86 percent of total public debt. Outstanding debt to all multilateral creditors stood at USD79 million (about 16 percent of GDP), or about 43 percent of the total external debt stock. The single largest creditor remains the Export-Import Bank of China (China Exim Bank), which accounts for 57 percent of total external debt stock (chart).[5] Tonga started repayments to China Exim Bank in FY2019 with larger repayments coming due starting in FY2024. The spike reflects both the previously extended grace period and the DSSI rescheduling. Tonga's request for a temporary suspension of its calendar year 2020 and 2021 debt service to the China Exim Bank under the G20 Debt Service Suspension Initiative (DSSI) has been accepted and the rescheduled debt repayments are included in the baseline scenario (text table). Under the DSSI, debt deferral amounts to 1.4 percent of GDP in FY2021 and 0.7 percent of GDP in FY2022.

Outstanding External PPG Debt by Creditor, June 2022
(in percent of total external PPG debt)



Source: Ministry of Finance.

Total Public and Publically Guaranteed Debt
(in percent of GDP)



Sources: Ministry of Finance; and IMF staff estimates.

Tonga: Repayment Schedule for Existing External Debt, FY2023–33

	FY2023	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032	FY2033
	(In percent of GDP)										
Total external debt	1.6	3.3	3.1	2.9	2.7	2.5	1.6	0.9	0.3	0.3	0.3
Multilateral	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.4	0.3	0.3	0.3
ADB	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.1	0.1
IDA/WB	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
IFAD[1/]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
IMF	0.0	0.0	0.0	0.0	0.3	0.4	0.5	0.5	0.4	0.2	0.1
Bilateral	1.1	2.8	2.6	2.4	2.3	2.0	1.2	0.5	0.0	0.0	0.0
EXIM Bank of China	1.1	2.8	2.6	2.4	2.3	2.0	1.2	0.5	0.0	0.0	0.0

1/ Repayments are around 0.03 percent of GDP per year.
Sources: Ministry of Finance; and IMF staff estimates.

4. Tonga's domestic debt obligations are relatively small. Public domestic debt stood at USD30 million (about 6.1 percent of GDP) at end-June 2022, accounting for 14 percent of total public debt.

[5] The loans from China Exim Bank are denominated in Chinese renminbi.

Domestic financial institutions hold about half of the total domestic debt with the rest held mainly by domestic pension funds. The authorities plan to issue domestic debt in FY2023 and beyond, if necessary.

BACKGROUND ON MACROECONOMIC FORECASTS

5. The assumptions in the baseline scenario are consistent with the macroeconomic framework presented in the Staff Report for the 2023 Article IV consultation. The baseline projections reflect the authorities' commitments to undertake reforms to enhance growth and achieve Tonga's climate resilience and SDGs under its Strategic Development Framework (SDF) 2015–2025, which comprise: enhancing trade facilitation and access to credit; building more climate-resistant public infrastructure as envisaged in the newly approved National Infrastructure Plan; boosting domestic revenues, including through tax administration reforms and rationalization of tax exemptions; and containing the public sector wage bill within the fiscal target while strengthening social protection and targeted support for the vulnerable.[6]

- **Real GDP growth** is projected at 1.6 percent on average during FY2023–33 (Text Table). The economy is set to recover with the reopening of the border and resumed activities in FY2023 and FY2024. Tourist arrivals, construction projects, and pent-up demand are expected to support growth. However, the rebound is limited by high inflation and labor shortages that may delay reconstruction and investment projects. Over the medium to long term, GDP growth is projected to gradually converge to a potential rate of 1.2 percent, underpinned by continued remittances and sizable government development expenditures. This low long-term growth rate reflects Tonga's remoteness, limited production base, and vulnerability to natural disasters. The risks to the growth projections are tilted to the downside, stemming from an abrupt global slowdown or recession, commodity price volatility, an acceleration of outward migration, and deepening geo-economic fragmentation that could affect Tonga through supply chain disruptions. Tonga is also highly vulnerable to natural disasters and the threat of rising sea levels. Given the importance of remittance and aid inflows to Tonga's economy, disruptions in correspondent banking relationships (CBRs) would have significant knock-on effects on the economy, the financial sector, and debt sustainability. On the upside, a faster-than-anticipated rebound in tourism can help support consumption, while higher capital spending on climate-resilient infrastructure projects could increase climate resilience and lift potential growth.

- **Natural disaster**: The baseline scenario incorporates the effect of natural disasters and climate change over the longer-term, as well as the recent progress in strengthening contingency planning through the ongoing implementation of the Tonga Disaster Risk Financing Strategy 2021 – 2025. The years FY2023–FY2025 are assumed to be disaster-free to simplify the policy discussion of the near-term outlook. From FY2026 onwards, the baseline incorporates the

[6] The socioeconomic impacts from the volcanic eruption and ashfalls and COVID-19 are expected to be disproportionately larger for households relying on agriculture and tourism, likely leading to higher overall poverty given Tonga's dependence on these sectors. Two-thirds of Tonga's households derive income from agriculture, and a third of the total households had at least one member working in tourism-linked sector before the global pandemic. Furthermore, the recent increase in commodity prices has pushed up the cost of living for all households, but the economic impact is expected to be especially acute for the poor and vulnerable.

average long-term effects of natural disasters and climate change by lowering annual GDP growth by 0.16 percentage points. These estimates are based on the findings of IMF staff analysis on the impact of natural disasters in Pacific Island countries.[7]

- **Inflation** is projected to average 4.3 percent during FY2023–33 (Text Table). Inflation is expected to peak in FY2023, reflecting both imported inflation and rising domestic prices due to the ongoing labor shortages. Afterwards, inflation is expected to gradually decline and converge to the steady state level of 3½ percent.[8]

- The **non-interest current account deficit** is estimated to have reached 7.5 percent of GDP in FY2023, owing to higher food and fuel prices and imports of reconstruction-related materials. In the medium to long term, the non-interest current account deficit is projected to average 7.7 percent of GDP over FY2023–FY2033, reflecting persistent weakness in export competitiveness, continued heavy reliance on imports to support domestic demand, a return of remittances to pre-pandemic levels as share of GDP, and a reduction in official transfer inflows after current commitments are met (Text Table). Beyond FY2033, the export-to-GDP ratio is expected to gradually increase, reflecting enhanced competitiveness and productivity in the service sector including tourism due to continued investment in climate resilience and infrastructure and trading partners' growth rates being higher than Tonga's.

- **Net FDI inflows** are expected to stand at 0.2 percent of GDP over FY2023–33.

[7] Lee, D., H. Zhang and C. Nguyen, 2018, "The Economic Impact of Natural Disaster in Pacific Island Countries: Adaptation and Preparedness", the IMF Working Paper No. 18/108.

[8] The upward revision of the long-term inflation forecast from the 2022 DSA mainly reflects the recent increase in the frequency of natural disasters in Tonga, which we expect to continue due to climate change. Over the period of 2015-2019, during which Tonga experienced a total of five natural disasters that led to supply disruptions, the annual average inflation rate recorded 3.3 percent.

Tonga: Baseline Macroeconomic Projections
(In percent of GDP, unless otherwise stated)

Current DSA	FY2023	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032	FY2033	FY2012-22 Historical average	FY2023-33 average
Real GDP growth (in percent)	2.6	2.5	2.2	1.6	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.0	1.6
GDP deflator in US dollars (change in percent)	7.2	3.6	2.3	2.1	2.1	2.2	2.0	2.2	2.1	2.1	2.1	-0.3	2.7
Inflation rate (GDP deflator, in percent)	10.0	5.8	4.1	3.4	3.2	3.4	3.2	3.4	3.5	3.5	3.5	2.7	4.3
Non-interest current account deficit	7.5	6.8	7.2	7.3	7.4	7.8	8.3	8.9	8.5	7.8	7.1	5.7	7.7
Net FDI (negative = inflow)	0.2	0.2	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.1	0.2
Primary deficit	-1.0	6.3	8.0	9.0	10.0	11.4	11.7	10.8	10.0	8.2	6.7	-2.5	8.3
Grants	24.4	17.0	12.9	11.1	9.5	7.5	6.5	5.1	4.5	4.0	4.0	17.4	9.7

Previous DSA	FY2023	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032	FY2033	FY2011-21 Historical average	FY2023-33 average
Real GDP growth (in percent)	3.2	3.0	2.5	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.3	2.1
GDP deflator in US dollars (change in percent)	3.7	3.4	2.6	2.0	1.5	1.6	2.2	2.6	2.6	2.6	2.4	0.2	2.5
Inflation rate (GDP deflator, in percent)	4.7	3.4	2.8	2.7	2.5	2.5	2.7	2.8	2.9	3.0	2.9	2.2	3.0
Non-interest current account deficit	22.7	19.3	8.1	13.8	13.8	16.1	9.9	10.6	9.9	9.6	9.0	5.6	13.0
Net FDI (negative = inflow)	1.2	0.8	1.0	1.0	0.9	-1.0	-1.0	-1.0	-1.2	-1.3	-1.2	1.3	-0.2
Primary deficit	4.2	4.4	8.1	8.8	10.2	11.7	12.5	11.7	11.5	11.9	6.5	-2.4	9.2
Grants	20.5	14.6	12.7	11.1	9.3	7.4	6.5	5.0	4.9	4.1	4.0	16.5	9.1

- **External borrowing**: Except for the IMF's second RCF-supported financing (disbursed in July 2022), new external borrowing is expected to commence in FY2025 and to gradually increase over the medium term to refinance debt repayments coming due and the primary deficit that is expected to reach double digits over FY2027–2031. Even if the authorities were to rely only on grants for their budget spending needs, under the baseline scenario, they would still need to borrow externally to meet their existing loan repayments while maintaining sufficient international reserves.[9] With the new external borrowing, the level of international reserves, currently estimated at 10.9 months of next year's imports at end-FY2023, is expected to remain stable in the coming years despite the large repayments to China EXIM Bank that start coming due beginning in FY2024 (annual payments of about 2.2 percent of GDP in FY2024–29 on average). The authorities will continue to seek new concessional and grant financing commitments from bilateral donors and international financial institutions to contain external debt and to refrain from non-concessional borrowing (consistent with Tonga's compliance with a non-concessional external borrowing ceiling as part of the World Bank's Sustainable Development Finance Policy, SDFP). This commitment is reflected in the government's medium-term debt strategy, which is being used to guide fiscal policy and debt management decisions. The new public financial management (PFM) legislation, currently under preparation, is also

[9] All lending are assumed for all years in the projection period for which grant finance has not already been committed, consistent with "Guidance Note on the Bank-Fund Debt Sustainability Framework for Low-income Countries" (http://www.imf.org/en/Publications/Policy-Papers/Issues/2018/02/14/pp122617guidance-note-on-lic-dsf). World Bank IDA financing will remain fully grants until FY2025, then a mix of grants and loans until FY2028, after which it is fully loans. The IDA loans are assumed to be on regular concessional credit terms for small economies. For other multilateral donors, new commitments from 2026 onwards are assumed to be on concessional loan terms. The terms are 40-year maturity, 10-year grace period, 0.75 percent service charge. The discount rate used to calculate the net PV of external debt is the default value set at 5 percent.

expected to improve debt management. Both the debt strategy and PFM legislation are supported by the World Bank's SDFP. [10]

- **Fiscal outlook:** The primary fiscal balance is expected to shift from 1 percent of GDP surplus in FY2023 to 6.3 percent of GDP deficit in FY2024 and then double-digit deficits during FY2027-FY2031 (Table). The surplus in FY2023 was mainly driven by a higher level of grants and slower than anticipated budget execution of reconstruction projects. The higher deficit in FY2024 reflects the normalization of grants from an exceptionally high level. The long-term deterioration is primarily due to lower grant inflows after existing commitments are met, as well as to sizable spending needs for covering infrastructure gaps and achieving climate resilience and SDGs. At the same time, domestic revenues are expected to improve over time, reflecting the authorities' ongoing efforts in revenue administration and tax arrears collection. Public sector wage bills are projected to drop below the fiscal target of 53 percent of domestic revenue in FY2024 thanks to recovering revenues and reforms to limit overtime pay. The fiscal deficit in percent of GDP is expected to decline after FY2029 as capital spending needs moderate following the completion of the authorities' multi-year reconstruction program and the accumulation of climate-resilient capital over the next decade.

6. The realism tools indicate that the primary balance projections are reasonable (Figure 4). The fiscal forecasts between FY2023 and FY2025 are not overly optimistic as the projected three-year adjustment lies in the left section of the distribution of past adjustments of the primary fiscal deficit (Figure 4). The real growth forecast for FY2023 reflects a strong rebound as the economy emerges from the twin shocks in FY2022. The realism of projections for public and private investment rates and their contribution to real GDP could not be calculated due to the unavailability of private investment data.

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TESTS

7. Tonga's debt-carrying capacity applied in the 2023 DSA remains strong. Tonga's Composite Indicator (CI) index, which has been calculated based on the April 2023 WEO and the 2021 CPIA is 3.06, indicating that the county's debt-carrying capacity remains strong according to the revised low-income country (LIC)-DSA framework.

8. Given the severity and frequency of natural disasters in Tonga, a tailored one-time stress test for natural disaster shocks was conducted.[11] As a small developing state prone to natural disasters,

[10] The Performance and Policy Action (PPA) for FY21 was cabinet approval of a medium-term debt strategy (covering fiscal years commencing FY21) which includes targets for external and domestic debt. The PFM bill is a PPA for FY22, FY23 and FY24, which outlines key fiscal responsibility principles to guide fiscal policy and a numerical limit on the stock of government guaranteed debt.

[11] All the standardized stress tests (i.e., growth shock, export shock, combination shock, historical scenario) are described in Table 8 of the new "Guidance Note on the Bank-Fund Debt Sustainability Framework for Low-income Countries" (http://www.imf.org/en/Publications/Policy-Papers/Issues/2018/02/14/pp122617guidance-note-on-lic-dsf).

Tonga is automatically subject to the standard natural disaster shock in the DSA. The Emergency Events Database (EM-DAT) shows that the country's largest damage from natural disasters during 1980–2016 was 28.2 percent of GDP. Since only part of the economic damage is expected to be financed by external debt, the DSA assumes a one-off shock of 14 percentage points (ppts) to the debt to-GDP ratio in FY2023. Real GDP growth and exports are lowered by 3 and 7 ppts, respectively, in the year of the shock (Lee et al., 2018).

Tonga: Composite Indicator and Threshold Tables

Calculation of the CI Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.442	1.33	43%
Real growth rate (in percent)	2.719	0.837	0.02	1%
Import coverage of reserves (in percent)	4.052	57.963	2.35	77%
Import coverage of reserves^2 (in	-3.990	33.597	-1.34	-44%
Remittances (in percent)	2.022	15.494	0.31	10%
World economic growth (in percent)	13.520	2.856	0.39	13%
CI Score			3.06	100%
CI rating			Strong	

Applicable thresholds

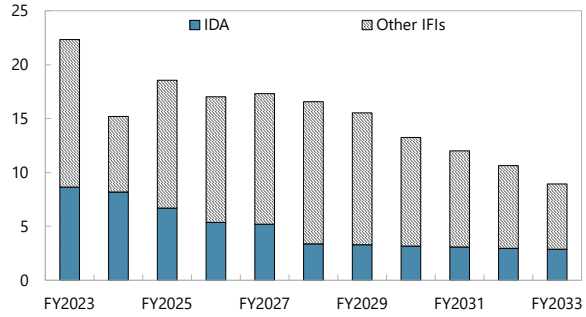
APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	240
GDP	55
Debt service in % of	
Exports	21
Revenue	23

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	70

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

9. Under the baseline scenario, PV of external debt-to-GDP ratio will breach its threshold starting from FY2033 (Figure 1 and Table 3). In the near term, the external debt-to-GDP ratio is expected to decline from 39.4 percent in FY2022 to 38 percent in FY2024, reflecting large repayments to China Exim Bank and the second RCF. From FY2025 onwards, Tonga will need to incur new external debt to help finance its large spending needs to achieve SDGs and climate resilience. As a result, external debt-to-GDP ratio will breach the authorities' fiscal anchor of 50 percent in FY2026. The PV of the external debt-to-GDP ratio is expected to breach the threshold of 55 percent starting from FY2033. The breach is not a single one-off event as external debt will continue to rise until FY2043 (Table 1). Under a

Projected External Net Financing Flows
(By official creditors, in percent of GDP)



Sources: IMF Staff Projections
Note: net financing flows are calculated as grants plus new debt disbursement minus repayment

combination of diverse shocks, which is the most extreme scenario, the PV of external debt-to-GDP ratio would breach the threshold earlier in FY2028.

10. The tailored natural disaster shock would significantly worsen the external debt path (Table 3). The PV of external debt-to-GDP ratio would jump up in FY2024 when the one-off shock is assumed, breach the threshold three years earlier in FY2030. Given recent experience, there is a significant possibility that multiple severe natural disasters could occur within a ten-year timeframe. Multiple natural disasters could have a larger cumulative negative effect on external debt sustainability due to larger reconstruction needs (which may require additional debt financing) and also by lowering the long-term growth.

PUBLIC DEBT SUSTAINABILITY ANALYSIS

11. Under the baseline scenario, the PV of the public debt-to-GDP ratio would breach 70 percent from FY2033 onwards (Figure 2). The public debt dynamic is driven by large primary fiscal deficit, as shown in Figure 3. Compared to the previous DSA, nominal public debt-to-GDP ratio is lower from FY2023 onward due to additional grants committed in FY2023 and FY2024.

12. The standardized sensitivity analysis indicates an earlier breach of debt threshold (Figure 2, Table 4).[12] The most extreme shock scenario is the growth shock, where growth rate is one standard deviation below the baseline for two years starting from 2023. The breach is not a single one-off event as public debt will continue to rise until FY2043 (Table 2). The PV of the public debt-to-GDP ratio would cross 70 percent of GDP in FY2029 under a multi-year GDP growth shock and breach the threshold in FY2030 under an exports shock.

13. The tailored one-time natural disaster shock worsens public debt. The PV of the public debt-to-GDP ratio would cross 70 percent of GDP in FY2030, three years earlier than in the baseline scenario.

14. A tailored stress test for the combined contingent liability shock also causes a deterioration in public debt sustainability. The trajectory of the PV of the public debt-to-GDP ratio shifts upwards by 4 percentage points from the baseline.

[12] Real GDP growth (in USD) is set to its historical average minus one standard deviation or the baseline projection minus one standard deviation, whichever is lower in the second and third years of the projection period.

RISK RATING AND VULNERABILITIES

15. Tonga's risk of external debt distress is assessed as high. The analysis using the LIC debt sustainability framework (DSF) indicates a "moderate" mechanical risk rating for external debt distress. However, under the baseline scenario, the PV of the external debt-to-GDP ratio is expected to breach its indicative threshold from FY2033 until FY2043, due to a sustained and persistent increase in the external debt ratio over this period because of high financing needs to address risks from climate change in the long term. Given the expected significant breach over this extended projection horizon, staff applied judgement and assessed the risk rating of external debt distress as "high."

16. The overall risk of debt distress is also assessed as high. The analysis using the LIC DSF indicates a "moderate" mechanical rating. However, the PV of the public debt-to-GDP ratio is projected to breach its indicative benchmark from FY2033 until FY2043, reflecting both large investment needs and decreasing capital grant commitments over this period.[13] Considering the expected significant breach over this extended projection horizon, together with the average long-term effects of natural disasters on Tonga's growth and fiscal position, staff assess the risk rating of the overall debt distress as "high."

17. In the short term, fiscal policy should prioritize reconstruction and protecting vulnerable groups, while containing public sector wages. Fiscal policy should focus on rebuilding and repairs while supporting vulnerable households and businesses amid high inflation. On the other hand, current spending, especially the recurrent portions such as public sector wages, should be contained to ease inflation pressures and preserve fiscal buffers.

18. Beyond the short term, staff assess that Tonga's public debt could be put on a sustainable path with feasible domestic fiscal measures and additional external donor support. Under an illustrative fiscal adjustment scenario, the PV of public debt-to-GDP ratio could stabilize below the 70 percent threshold over the long term, even with a multi-year growth shock, while the PV of external debt-to-GDP ratio would stabilize below the 55 percent benchmark under this scenario. The fiscal adjustments could take place post-reconstruction and include a gradual increase of tax revenues and reduction of current expenditure to levels similar to other PICs. New grant commitments consistent with the historical trends are also essential under this fiscal path to fund capital investment projects for SDGs and climate resilience.[14] The government's strategy to further improve revenue administration, collect tax arrears, contain public wage bill within the fiscal target, and pursue additional grants from development partners and avoid any new non-concessional borrowing from external creditors is aligned with this fiscal adjustment path and feasible considering Tonga's strong policy track record in the run up to the global pandemic.

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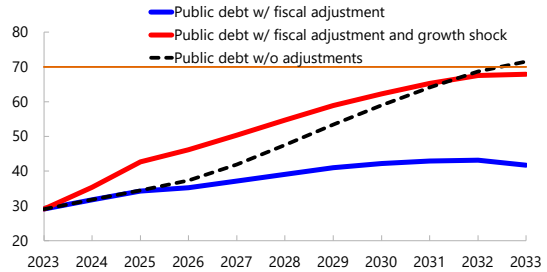
[13] Cash and in-kind grants averaged 16.7 percent of GDP annually over FY2015–2019. Under the baseline, staff assume continued budget support in line with historical levels and capital grants falling to zero from FY2030 onwards.

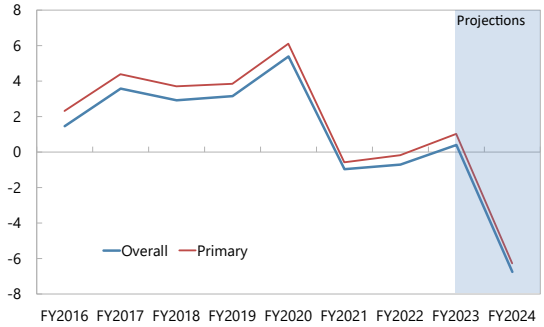
[14] The amount of grants declines over time in the baseline scenario since it only includes committed or highly likely grants, in line with the LIC DSA guidance note. Given the authorities' prudent fiscal policies in the past and various reform initiatives supported by development partners, public debt can feasibly be put on a sustainable path.

Public Debt with Fiscal Adjustment
(Present value of debt in percent of GDP)



Source: IMF staff estimates.
Note: The most extreme shock scenario is the growth shock. The red line does not include the public investment reforms which would lower the debt path further.

Overall and Primary Balance
(in percent of GDP)



Sources: Authorities and Staff Projections

Alternative Scenario with Fiscal Adjustment
(In percentage points of GDP, deviation from baseline)

	FY2025	FY2026	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032	FY2033
Domestic fiscal measures	0.8	1.6	2.4	3.2	3.7	3.7	3.7	3.7	3.7
Better VAT compliance 1/	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Less tax exemptions 2/		0.5	1.0	1.5	2.0	2.0	2.0	2.0	2.0
Lower G&S			-0.3	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6
Lower public sector wage bill	-0.3	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6
Additional Grants		2.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Change in primary balance	0.8	3.6	5.4	6.2	6.7	6.7	6.7	6.7	6.7
Impact on GDP growth 3/									
Without additional reform	-0.3	-0.3	-0.1	-0.1	0.0	0.0	0.0	0.0	0.0
With additional reform 4/	-0.3	-0.2	0.0	0.1	0.2	0.3	0.4	0.4	0.5
Impact on inflation 5/									
Without additional reform	-0.2	-0.2	-0.1	-0.1					
With additional reform 4/	-0.2	-0.2	-0.1	-0.1					
Memo. (in percent of GDP)									
Projected primary balance w/adjustment	-7.2	-5.4	-4.6	-5.2	-5.0	-4.1	-3.3	-1.5	0.0

1/ For reference, World Bank (2023) estimates the potential revenue impact from such reform for Tonga at about 6.2 percent of GDP.
2/ For reference, World Bank (2023) estimates the potential revenue impact from reducing exemptions and zero ratings at about 3.2 percent of GDP.
3/ The growth impact associated with fiscal adjustment is based on an average multiplier value of 0.2-0.4 over the projection period, comparable with estimates for other emerging market and low-income countries in the literature. The growth impact associated with additional reform is obtained from the IMF's DIGNAD model calibrated to the Tongan economy and in line with other studies (e.g., Solomon Islands, 2018 Article IV consultation, Selected Issues Paper, Chapter 2).
4/ The reform, assumed to be implemented over FY2024-FY2032, enhances Tonga's public investment efficiency from 40 percent to 60 percent. The starting value of 40 percent corresponds to the median for a sample of emerging market and low-income countries in Dabla-Norris and others (2011) and the increase of 20 percentage points is equivalent to about one standard deviation.
5/ The impact on inflation under the "with additional reform" scenario is driven by the demand-dampening effects of fiscal consolidation. While the assumed increase in public investment efficiency (the rate at which a given amount of public investment is converted to public capital) would help boost Tonga's productive capacity, thereby helping ease inflationary pressures, the accompanying demand-stimulating effects (e.g., crowding-in of private investment) could offset these supply-side effects. Given the large uncertainties surrounding the long-term impacts on inflation, we do not show estimated impacts beyond FY2028 to avoid conveying a false sense of precision.

AUTHORITIES' VIEWS

The authorities agreed on the need for fiscal adjustment and emphasized the importance of sustained support from development partners. They shared staff's concern regarding the high public sector wage bill relative to the narrow domestic revenue base and are taking necessary steps to contain it, including by limiting overtime allowances and undertaking a comprehensive

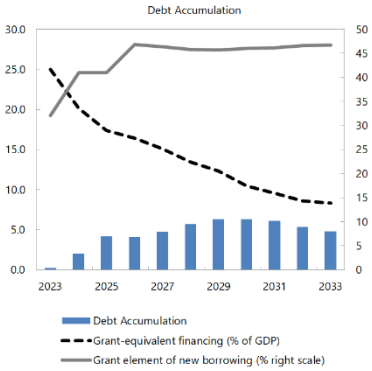
organizational review of the government. However, growing pay differentials with New Zealand and Australia in a context of strong labor demand there have made it difficult to retain civil servants, thereby exerting upward pressure on public sector wages. The authorities underscored their intention to seek new grant financing from development partners to meet climate resilience and development goals, while ensuring debt remains on a sustainable path.

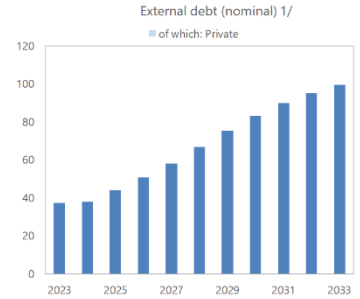
Table 1. Tonga: External Debt Sustainability Framework, Baseline Scenario, FY2020–2043

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2033	2043	Historical	Projections
External debt (nominal) 1/	**37.4**	**41.2**	**39.4**	**37.2**	**38.0**	**43.9**	**50.7**	**58.2**	**66.7**	**99.3**	**99.9**	**40.7**	**67.0**
of which: public and publicly guaranteed (PPG)	**37.4**	**41.2**	**39.4**	**37.2**	**38.0**	**43.9**	**50.7**	**58.2**	**66.7**	**99.3**	**99.9**	**40.7**	**67.0**
Change in external debt	1.0	3.9	-1.8	-2.2	0.7	5.9	6.8	7.5	8.5	4.2	0.2		
Identified net debt-creating flows	**7.6**	**6.9**	**4.6**	**7.2**	**6.4**	**6.9**	**7.1**	**7.4**	**7.8**	**7.1**	**4.7**	**7.3**	**7.6**
Non-interest current account deficit	**4.8**	**5.1**	**6.0**	**7.5**	**6.8**	**7.2**	**7.3**	**7.4**	**7.8**	**7.1**	**4.7**	**5.7**	**7.7**
Deficit in balance of goods and services	43.3	49.9	53.9	63.0	54.8	52.7	51.2	50.2	48.9	45.5	43.7	**44.1**	**50.6**
Exports	20.6	12.4	11.9	16.6	19.0	20.2	21.2	21.8	22.3	24.5	28.1		
Imports	63.9	62.3	65.8	79.6	73.8	72.8	72.4	72.0	71.2	70.0	71.8		
Net current transfers (negative = inflow)	-30.3	-38.7	-42.8	-50.5	-42.2	-39.1	-37.0	-35.3	-33.1	-30.2	-30.5	**-33.5**	**-35.7**
of which: official	-4.1	-2.2	-5.3	-15.6	-12.9	-12.9	-11.1	-9.5	-7.5	-4.0	-4.0		
Other current account flows (negative = net inflow)	-8.2	-6.0	-5.1	-5.0	-5.8	-6.4	-6.9	-7.4	-8.0	-8.2	-8.4		
Net FDI (negative = inflow)	**0.7**	**0.1**	**0.5**	**0.2**	**0.2**	**0.3**	**0.2**	**0.2**	**0.2**	**0.2**	**0.1**	**1.1**	**0.2**
Endogenous debt dynamics 2/	**2.1**	**1.7**	**-1.9**	**-0.6**	**-0.6**	**-0.6**	**-0.4**	**-0.2**	**-0.2**	**-0.2**	**-0.2**		
Contribution from nominal interest rate	0.5	0.2	0.4	0.4	0.3	0.2	0.3	0.4	0.5	0.9	1.0		
Contribution from real GDP growth	-0.2	1.0	0.8	-0.9	-0.9	-0.8	-0.7	-0.6	-0.7	-1.1	-1.2		
Contribution from price and exchange rate changes	1.7	0.5	-3.0	...	...	...	...	...	...	...	...		
Residual 3/	**-6.7**	**-3.0**	**-6.4**	**-9.4**	**-5.7**	**-1.0**	**-0.4**	**0.1**	**0.7**	**-2.8**	**-4.5**	**-8.1**	**-2.1**
of which: exceptional financing	0.0	-1.4	-0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**23.8**	**21.9**	**22.5**	**25.5**	**28.6**	**32.2**	**36.6**	**57.1**	**58.0**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**199.9**	**132.1**	**118.6**	**126.5**	**135.1**	**147.7**	**164.5**	**232.9**	**206.6**		
PPG debt service-to-exports ratio	**7.4**	**5.9**	**12.8**	**12.2**	**19.2**	**17.1**	**15.9**	**14.9**	**13.8**	**11.9**	**16.7**		
PPG debt service-to-revenue ratio	**6.0**	**2.8**	**5.8**	**8.7**	**14.8**	**14.0**	**13.5**	**13.0**	**12.3**	**11.2**	**16.5**		
Gross external financing need (Million of U.S. dollars)	34.6	27.8	39.7	53.4	62.1	66.4	68.6	71.1	75.0	81.1	105.1		
Key macroeconomic assumptions													
Real GDP growth (in percent)	0.5	-2.7	-2.0	2.6	2.5	2.2	1.6	1.2	1.2	1.2	1.2	**1.0**	**1.6**
GDP deflator in US dollar terms (change in percent)	-4.6	-1.4	7.9	7.2	3.6	2.3	2.1	2.1	2.2	2.1	2.1	**-0.3**	**2.7**
Effective interest rate (percent) 4/	1.4	0.4	0.9	1.0	0.7	0.7	0.7	0.8	0.8	1.0	1.0	**1.4**	**0.8**
Growth of exports of G&S (US dollar terms, in percent)	-10.2	-42.2	1.6	53.0	21.7	11.2	8.8	6.6	5.7	5.7	4.6	**-0.5**	**12.1**
Growth of imports of G&S (US dollar terms, in percent)	-4.2	-6.5	11.8	33.1	-1.6	3.2	3.1	2.8	2.3	3.1	4.1	**1.6**	**5.3**
Grant element of new public sector borrowing (in percent)			20.1	32.1	41.0	41.0	46.8	46.3	45.8	46.7	46.4	**...**	**44.0**
Government revenues (excluding grants, in percent of GDP)	25.1	25.9	26.2	23.3	24.6	24.7	24.8	24.9	25.0	26.3	28.4	**23.5**	**25.2**
Aid flows (in Million of US dollars) 5/	512.0	517.6	493.7	511.2	451.7	394.4	352.0	300.6	249.0	145.2	67.6	**...**	
Grant-equivalent financing (in percent of GDP) 6/	19.1	22.4	19.0	25.0	20.1	17.3	16.4	15.1	13.5	8.3	7.3	**...**	**14.2**
Grant-equivalent financing (in percent of external financing) 6/	100.0	100.0	100.0	95.5	81.7	73.1	72.9	70.1	65.7	62.7	65.7	**...**	**70.2**
Nominal GDP (Million of US dollars)	490	470	498	547	581	607	630	652	674	794	1,100		
Nominal dollar GDP growth	-4.1	-4.0	5.8	10.0	6.2	4.5	3.8	3.4	3.4	3.3	3.3	**0.7**	**4.4**
Memorandum items:													
PV of external debt 7/	...	...	23.8	21.9	22.5	25.5	28.6	32.2	36.6	57.1	58.0		
In percent of exports	...	...	199.9	132.1	118.6	126.5	135.1	147.7	164.5	232.9	206.6		
Total external debt service-to-exports ratio	7.4	5.9	12.8	12.2	19.2	17.1	15.9	14.9	13.8	11.9	16.7		
PV of PPG external debt (in Million of US dollars)			118.5	119.7	130.8	155.1	180.2	210.0	247.1	453.7	638.2		
(PVt-PVt-1)/GDPt-1 (in percent)				0.3	2.0	4.2	4.1	4.7	5.7	4.8	2.0		
Non-interest current account deficit that stabilizes debt ratio	3.8	1.2	7.8	9.7	6.1	1.3	0.5	-0.1	-0.7	2.9	4.5		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**Yes**





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα(1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

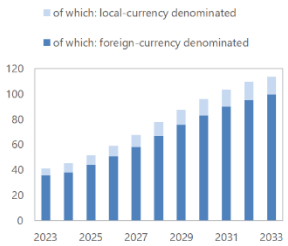
Table 2. Tonga: Public Sector Debt Sustainability Framework, Baseline Scenario, FY2020–2043

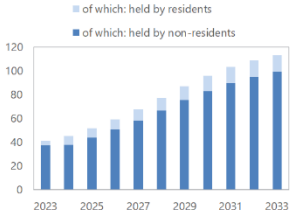
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2033	2043	Historical	Projections
Public sector debt 1/	**42.5**	**47.8**	**45.4**	**41.2**	**45.4**	**51.6**	**59.1**	**67.6**	**77.5**	**113.3**	**110.4**	**46.4**	**77.4**
of which: external debt	**37.4**	**41.2**	**39.4**	**37.2**	**38.0**	**43.9**	**50.7**	**58.2**	**66.7**	**99.3**	**99.9**	**40.7**	**67.0**
Change in public sector debt	1.2	5.3	-2.4	-4.1	4.1	6.2	7.5	8.6	9.8	4.1	-0.4		
Identified debt-creating flows	**-3.5**	**2.2**	**-0.9**	**-2.8**	**4.8**	**6.6**	**7.8**	**8.9**	**10.2**	**4.8**	**1.6**	**-2.2**	**6.8**
Primary deficit	**-6.1**	**0.6**	**0.2**	**-1.0**	**6.3**	**8.0**	**9.0**	**10.0**	**11.4**	**6.7**	**1.9**	**-2.5**	**8.3**
Revenue and grants	44.2	48.3	45.2	47.7	41.6	37.6	35.9	34.5	32.5	30.3	32.4	**41.0**	**34.9**
of which: grants	19.1	22.4	19.0	24.4	17.0	12.9	11.1	9.5	7.5	4.0	4.0		
Primary (noninterest) expenditure	38.1	48.9	45.4	46.7	47.9	45.7	44.9	44.5	44.0	36.9	34.3	**38.5**	**43.2**
Automatic debt dynamics	**2.6**	**1.6**	**-1.1**	**-1.8**	**-1.5**	**-1.4**	**-1.2**	**-1.1**	**-1.2**	**-1.8**	**-0.3**		
Contribution from interest rate/growth differential	0.1	1.0	0.4	-1.8	-1.5	-1.4	-1.2	-1.1	-1.2	-1.8	-0.3		
of which: contribution from average real interest rate	0.3	-0.2	-0.6	-0.6	-0.5	-0.4	-0.4	-0.4	-0.4	-0.5	1.0		
of which: contribution from real GDP growth	-0.2	1.2	1.0	-1.1	-1.0	-1.0	-0.8	-0.7	-0.8	-1.3	-1.3		
Contribution from real exchange rate depreciation	2.5	0.6	-1.5	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**4.7**	**3.1**	**-1.5**	**-1.4**	**-0.7**	**-0.4**	**-0.3**	**-0.3**	**-0.3**	**-0.7**	**-2.1**	**1.2**	**-0.6**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	...	...	**30.1**	**26.1**	**30.1**	**33.4**	**37.2**	**41.9**	**47.7**	**71.5**	**68.8**		
PV of public debt-to-revenue and grants ratio	...	...	**66.5**	**54.7**	**72.5**	**88.8**	**103.5**	**121.5**	**146.5**	**236.2**	**212.0**		
Debt service-to-revenue and grants ratio 3/	**7.2**	**6.0**	**7.3**	**7.0**	**10.1**	**14.1**	**12.6**	**21.5**	**20.0**	**28.1**	**28.3**		
Gross financing need 4/	-2.9	3.5	3.5	2.3	10.5	13.3	13.6	17.4	17.9	15.2	11.1		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	0.5	-2.7	-2.0	2.6	2.5	2.2	1.6	1.2	1.2	1.2	1.2	**1.0**	**1.6**
Average nominal interest rate on external debt (in percent)	1.4	0.4	0.9	1.0	0.7	0.7	0.7	0.8	0.8	1.0	1.0	**1.5**	**0.9**
Average real interest rate on domestic debt (in percent)	7.5	5.4	-4.5	-7.2	-3.7	-1.4	-0.5	-0.2	-0.2	0.0	0.4	**-0.5**	**-1.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	7.0	1.6	-3.5	...	...	...	...	...	...	...	...	**1.8**	**...**
Inflation rate (GDP deflator, in percent)	-4.2	-1.9	8.5	10.0	5.8	4.1	3.4	3.2	3.4	3.5	3.1	**2.7**	**4.3**
Growth of real primary spending (deflated by GDP deflator, in percent)	1.1	25.0	-8.9	5.5	5.0	-2.5	0.0	0.2	0.1	-2.6	4.0	**6.5**	**-0.3**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-7.3	-4.7	2.6	3.1	2.1	1.8	1.6	1.4	1.6	2.6	2.3	**-3.1**	**2.1**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The general government, central bank . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

"Other debt creating or reducing flow" is the net acquisition of financial assets.

Table 3. Tonga: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, FY2023–2033

(In percent)

	Projections 1/										
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
PV of debt-to GDP ratio											
Baseline	22	23	26	29	32	37	42	46	51	54	**57**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	22	24	27	30	34	38	43	47	52	**56**	**59**
B. Bound Tests											
B1. Real GDP growth	22	24	29	32	37	42	47	52	**57**	**61**	**65**
B2. Primary balance	22	24	28	32	35	40	45	50	54	**58**	**61**
B3. Exports	22	27	38	41	45	50	**55**	**60**	**65**	**68**	**71**
B4. Other flows 3/	22	31	41	44	48	53	**58**	**62**	**67**	**70**	**72**
B5. Depreciation	22	28	23	27	32	37	43	49	54	**59**	**63**
B6. Combination of B1-B5	22	37	45	49	53	**58**	**64**	**69**	**74**	**78**	**80**
C. Tailored Tests											
C1. Combined contingent liabilities	22	25	28	32	36	40	45	50	55	**58**	**61**
C2. Natural disaster	22	29	33	38	42	47	53	**58**	**63**	**67**	**71**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio											
Baseline	132	119	127	135	148	164	184	201	216	226	233
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	132	124	134	143	156	172	191	206	221	232	**243**
B. Bound Tests											
B1. Real GDP growth	132	119	127	135	148	164	184	201	216	226	233
B2. Primary balance	132	125	139	149	163	180	199	216	231	**241**	**247**
B3. Exports	132	222	**411**	**426**	**453**	**490**	**534**	**573**	**606**	**626**	**634**
B4. Other flows 3/	132	164	204	209	220	236	**255**	**271**	**285**	**292**	**294**
B5. Depreciation	132	119	92	102	115	132	152	170	185	195	204
B6. Combination of B1-B5	132	**246**	198	**356**	**375**	**403**	**437**	**465**	**490**	**501**	**505**
C. Tailored Tests											
C1. Combined contingent liabilities	132	134	141	152	164	182	201	218	233	**243**	**249**
C2. Natural disaster	132	159	167	183	198	218	**241**	**261**	**278**	**290**	**298**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio											
Baseline	12	19	17	16	15	14	11	10	9	11	12
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	12	20	19	18	17	16	13	12	11	13	15
B. Bound Tests											
B1. Real GDP growth	12	19	17	16	15	14	11	10	9	11	12
B2. Primary balance	12	19	17	16	15	14	12	10	10	11	13
B3. Exports	12	**30**	**40**	**39**	**36**	**34**	**28**	**24**	**24**	**29**	**35**
B4. Other flows 3/	12	19	18	17	16	15	13	11	11	14	17
B5. Depreciation	12	19	17	15	14	13	11	9	9	10	10
B6. Combination of B1-B5	12	**27**	**32**	**30**	**28**	**26**	**22**	19	18	**26**	**29**
C. Tailored Tests											
C1. Combined contingent liabilities	12	19	17	16	15	14	12	10	10	11	12
C2. Natural disaster	12	20	19	17	17	15	13	11	11	12	13
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio											
Baseline	9	15	14	14	13	12	10	9	9	10	11
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	9	16	15	15	15	14	12	10	10	12	14
B. Bound Tests											
B1. Real GDP growth	9	16	16	15	15	14	12	10	10	11	13
B2. Primary balance	9	15	14	14	13	13	10	9	9	10	12
B3. Exports	9	15	15	15	15	14	12	10	10	12	15
B4. Other flows 3/	9	15	15	15	14	13	11	10	10	13	16
B5. Depreciation	9	19	18	16	16	15	12	10	10	12	11
B6. Combination of B1-B5	9	16	17	17	16	15	13	11	11	15	18
C. Tailored Tests											
C1. Combined contingent liabilities	9	15	14	14	13	13	11	9	9	10	11
C2. Natural disaster	9	15	14	14	14	13	11	9	9	11	12
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	23	23	23	23	23	23	23	23	23	23	23
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	44	44	44	44	44	44	44	44	44	44	**44**

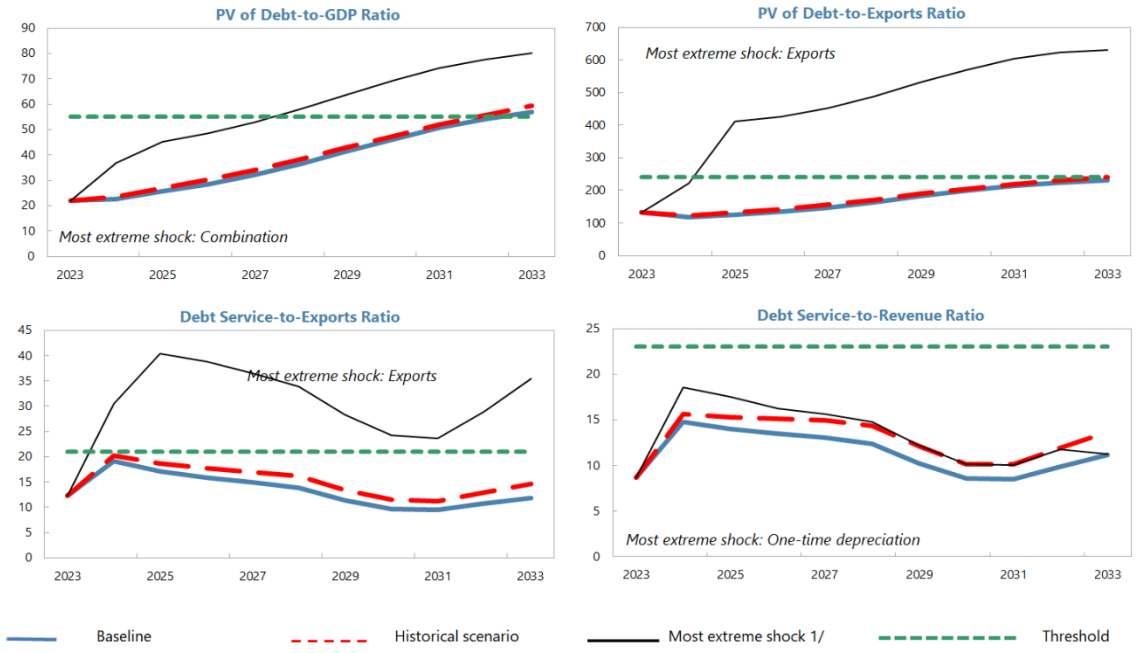
Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Tonga: Sensitivity Analysis for Key Indicators of Public Debt, FY2023–2033
(In percent)

	Projections 1/										
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
PV of Debt-to-GDP Ratio											
Baseline	26.1	30.1	33.4	37.2	41.9	47.7	53.7	59.4	64.6	68.7	**71.5**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	26	26	25	22	20	18	16	13	11	9	6
B. Bound Tests											
B1. Real GDP growth	26	33	41	47	55	63	**72**	**81**	**89**	**95**	**100**
B2. Primary balance	26	32	38	41	46	51	57	63	68	**72**	**75**
B3. Exports	26	34	44	48	52	58	64	**70**	**76**	**79**	**82**
B4. Other flows 3/	26	39	49	53	58	64	70	**76**	**81**	**85**	**87**
B5. Depreciation	26	34	34	34	36	38	41	44	47	49	50
B6. Combination of B1-B5	26	31	35	38	43	48	55	60	66	70	**73**
C. Tailored Tests											
C1. Combined contingent liabilities	26	35	38	41	46	52	58	63	69	**73**	**76**
C2. Natural disaster	26	41	45	48	53	59	66	**72**	**78**	**83**	**86**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	70	70	70	70	70	70	70	70	70	70	70
PV of Debt-to-Revenue Ratio											
Baseline	55	72	89	103	121	146	169	190	211	228	236
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	55	63	64	60	56	53	48	42	36	29	21
B. Bound Tests											
B1. Real GDP growth	55	78	105	127	153	189	221	253	283	310	326
B2. Primary balance	55	78	100	114	132	158	181	202	223	240	248
B3. Exports	55	82	117	133	152	179	203	225	246	264	270
B4. Other flows 3/	55	94	131	147	167	196	220	242	264	281	286
B5. Depreciation	55	85	93	98	106	119	132	143	155	165	168
B6. Combination of B1-B5	55	76	93	105	123	149	172	193	214	231	240
C. Tailored Tests											
C1. Combined contingent liabilities	55	84	101	115	133	159	182	203	224	241	250
C2. Natural disaster	55	97	117	132	153	181	207	231	254	274	285
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	7	10	14	13	22	20	16	23	23	22	28
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	7	10	14	13	22	20	16	23	23	20	25
B. Bound Tests											
B1. Real GDP growth	7	10	15	15	26	25	22	29	30	29	36
B2. Primary balance	7	10	14	15	24	21	17	23	24	23	29
B3. Exports	7	10	14	13	22	21	17	23	24	23	31
B4. Other flows 3/	7	10	15	13	22	21	17	24	24	25	32
B5. Depreciation	7	11	17	15	23	22	18	24	24	23	29
B6. Combination of B1-B5	7	10	14	13	22	20	16	23	23	22	28
C. Tailored Tests											
C1. Combined contingent liabilities	7	10	14	17	22	21	17	23	23	22	28
C2. Natural disaster	7	10	15	23	23	24	18	26	25	24	30
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

Figure 1. Tonga: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, FY2023–2033



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	No	
Natural disaster	Yes	Yes
Commodity price 2/	n.a.	n.a.
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.1%	1.1%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	28	28
Avg. grace period	7	7

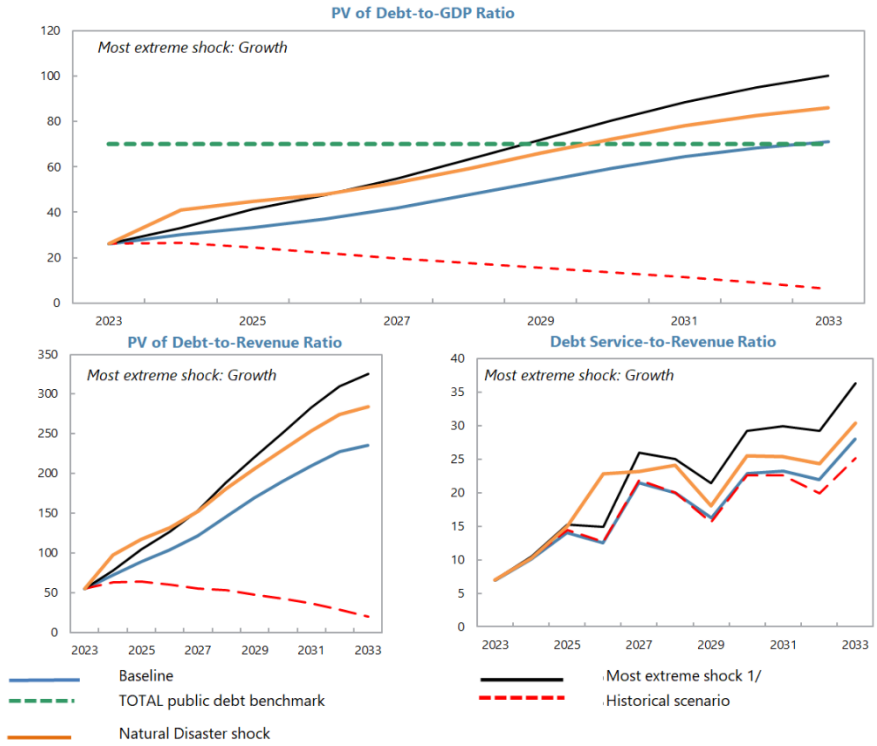
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2033. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Tonga: Indicators of Public Debt Under Alternative Scenarios, FY2023–2033



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	74%	74%
Domestic medium and long-term	26%	26%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.1%	1.1%
Avg. maturity (incl. grace period)	28	28
Avg. grace period	7	7
Domestic MLT debt		
Avg. real interest rate on new borrowing	-0.8%	-0.8%
Avg. maturity (incl. grace period)	2	2
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	0.0%	0.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

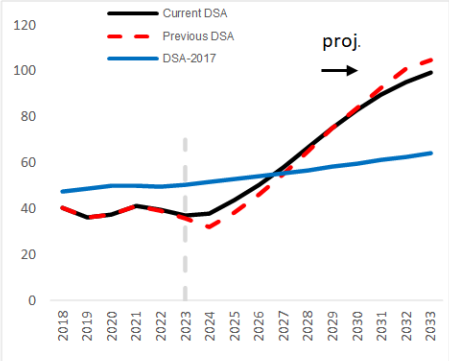
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2033. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

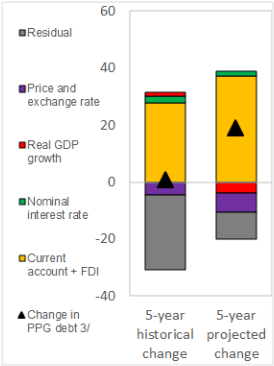
Figure 3. Tonga: Drivers of Debt Dynamics—Baseline Scenario External Debt

External Debt

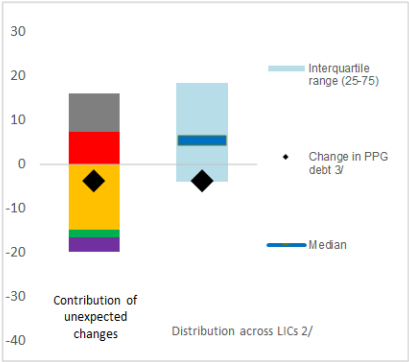
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

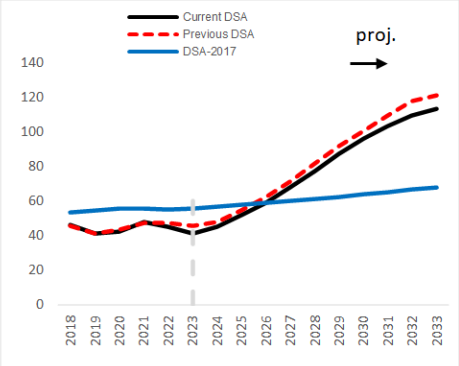


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

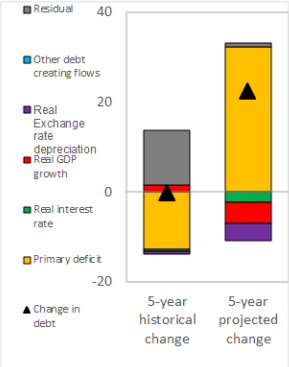


Public Debt

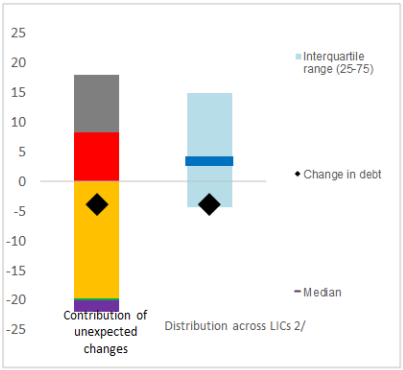
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



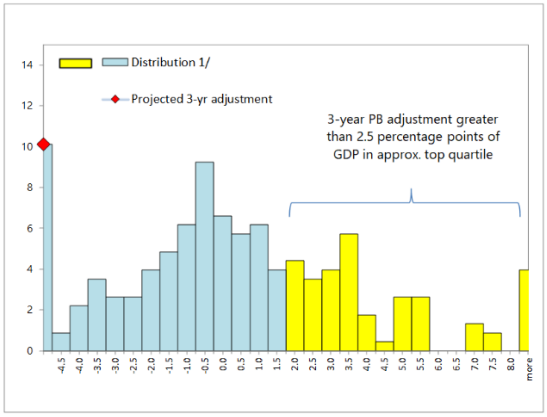
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

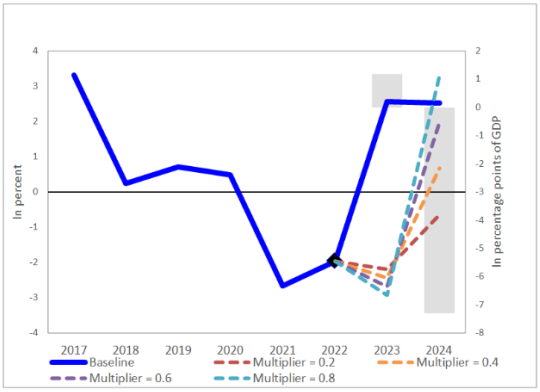
Figure 4. Tonga: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).